

Ethical Considerations for a Skincare Selection Machine Learning Model

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June 11, 2024

Introduction

The significance of appearance and perceived beauty has been widely acknowledged for its impact on psychological well-being and the ability to create a favorable impression on others (Blanco-Dávila, 2000; Georgievskaya, 2023). The skincare industry, a segment of the beauty and personal care market, specializes in developing, producing, and marketing products aimed at enhancing the appearance and health of the skin (Georgievskaya, 2023). This is a multi-billion dollar industry worldwide, and is predicted to continue to grow (Boon et al, 2020).

In this continuously growing skincare market, facial skincare products remain the most popular choice (Li et al, 2020). However, with thousands of products and ingredients available, consumers often feel overwhelmed, confused, and fatigued by the vast amounts of options available for their skincare product selections (Truswell, 2020). This is further exacerbated by the fact that skin composition and characteristics vary so widely, so skincare products which are very beneficial for some people can be unhelpful or even damaging to others (Truswell, 2020). Therefore, trying new products can be a daunting task for consumers due to fear of causing skin irritation from poor tolerance of a certain ingredient, or from adverse interactions with other skincare products.

It is important to consider that there are many skincare concerns, and certain ingredients are better suited for differing skin care concerns. Common facial skin concerns include acne, wrinkles, dryness, oiliness, dark spots/hyperpigmentation, rosacea, and enlarged pores (Li et al, 2020). Different types of skincare products (e.g. cleansers, toners, serums, creams, sunscreens) can address one or multiple concerns with strategic use of popular skincare ingredients (e.g. retinoids for skin cell turnover to

reduce wrinkles, chemical exfoliants to unclog pores, humectants to moisturize, antioxidants to boost sunscreen effectiveness and reduce hyperpigmentation, ceramides to improve skin barrier, SPF to prevent further wrinkles and sunspots, etc (Santilli & Jacoby, 2022).

The goal of this data science effort is for a machine learning model to enhance the process of customers selecting skincare products tailored to individual needs and concerns by analyzing product ingredient information, skincare expert review, combined with consumer data patterns to provide personalized recommendations and ease the customer's experience. This model is intended to be used by individual companies who sell a variety of skincare products.

Data Collection

The model will analyze data from the company's website. This will include basic product details (e.g. name, brand, size), ingredients list for each product (e.g. active and inactive ingredients), category for each product (e.g. cleanser, toner, serum, cream, sunscreen), and customer feedback (e.g. 5-star rating system and comments).

Expert opinion and review by dermatologists and skincare experts will be used to initially develop guidelines as to which ingredients are beneficial and harmful for which concerns. For instance, ingredients classified as retinoids are effective for wrinkles and hyperpigmentation. Ingredients classified as alpha-hydroxy acids (AHAs) or beta-hydroxy acids (BHAs) are effective chemical exfoliants to address acne (Santilli & Jacoby, 2022). However, it is commonly said that retinoids should not be mixed with chemical exfoliants in the same routine due to high risk of skin irritation which could lead

to a disruption of the skin barrier (Santilli & Jacoby, 2022). Similarly, many believe that retinoids can worsen pre-existing rosacea (Santilli & Jacoby, 2022). Therefore, dermatologists and skincare experts would need to guide initial structuring of the model so as to know which ingredients are beneficial for which concerns, which ingredients may be harmful for which concerns, and which ingredients to avoid mixing together in the same routine. This information will be obtained from an agency or team of dermatologists and qualified estheticians who are partnered with the company to provide this initial and ongoing information.

Potential customers who use a company's skincare website to browse and potentially order skincare products will engage in a brief pre-purchase questionnaire. The user will select their age range (e.g. 18-29, 30-44, 45-60, 60+), their skin tone (e.g. fair, medium, dark), their skin type (e.g. dry, normal, oily, combination), any known allergies or sensitivities, and will select their primary skincare concerns (e.g. acne, wrinkles, discoloration/hyperpigmentation, rosacea, enlarged pores, etc). The model will then output the top product recommendation for each major skincare category (i.e. cleanser, toner, serum, cream, sunscreen) to best meet the customer's needs. This pre-purchase questionnaire will ask for the customer's email address so that the company can send a post-purchase request for feedback following 1-2 months of a confirmed order containing all or some of the recommended products. The customer will be offered a small incentive by the company (e.g. 10% off discount code for the next order) if the customer provided feedback regarding their product. This can be done via a 5-star rating system and/or comment.

In sum, the collection of data relating to product information, skincare expert opinions, existing customer feedback, pre-purchase questionnaire, and post-purchase feedback will be used to train and test the model.

One ethical concern with the data collection phase is the issue of fairness with skin tones. This is a pervasive issue in the skincare industry, with a known history of bias and overrepresentation of lighter skin tones. The majority of cosmetic skincare studies are conducted on people with light skin tones, and with significant underrepresentation of people with darker skin tones, even as low as 4% of clinical trials being tested on brown or black skin (Georgievskaya, 2023; Sachdeva, 2009). To address this, the data collection phase will take special consideration for ingredients which may be less effective or even adverse to people with darker skin tones (Levy, 2019). This can also be mitigated by seeking out expert opinions from dermatologists and estheticians who are experienced in working with people of color for their skincare goals, and/or who are people of color themselves.

A second ethical concern is for privacy and data security. Collecting sensitive personal data, such as skin conditions, health history, and preferences, can lead to privacy violations if not handled properly. There is a risk of data breaches exposing this sensitive information. This can be mitigated by anonymous credentials in which the model will only relate to the skincare information and eliminate any personally identifiable information. The pre-purchase questionnaire will only have an email address as a potential identifier in order to have a means to send the post-purchase email. Data minimalism will be implemented, whereby the model will not ask for information related to names, addresses, or payment information, although these will be asked by the

company hosting this model to complete the purchase. Data encryption will also be utilized, in which data is encrypted both in transit and at rest to protect it from unauthorized access. Access controls will also be put in place to restrict who can view or modify the collected data.

A third ethical concern is for informed or explicit consent to data usage. Users may not be fully aware of what data is being collected, how it may be used, and who may have access to it. The company will implement informed and explicit consent. When a user leaves a review of a product, engages in the pre-purchase questionnaire and/or post-purchase email feedback, the user will be made aware that their feedback is being used to enhance the product selection experience for customers like them, and that all data will be anonymous. An opt-out button will be made easily available, and a portal for asking any further questions to the company regarding this data tracking system will also be readily available to promote bidirectional communication.

Analysis/Modeling

A major ethical concern in the analysis phase that must be considered is health and safety. The model must take very careful precaution to avoid recommendations that are likely to cause adverse skin reactions. This is why the model will integrate dermatologists' and skincare experts' guidelines into the recommendation engine to prioritize products based on ingredient suitability for specific skin concerns and tones, and exclude products which have been shown to be not effective or even harmful for specific skin concerns and skin tones. The experts' recommendations will also be crucial to avoid adverse interactions between products, such as retinoids and chemical exfoliants as previously mentioned. Interactions due to duplications will also need to be

avoided per expert opinion. For example, the model should not recommend both a retinoid serum and a retinoid cream in the same regimen, as retinoids can be highly irritating to the skin if used in excess (Santilli & Jacoby, 2022).

The model will use a rule-based system to specify which products are contraindicated in which situations, and logical conditions in the model can exclude redundant active ingredients. The rule-based system can further adjust recommendations dynamically. In this way, poor recommendations will be filtered out prior to presenting the final list to the user. Additionally, the model will take weight to post-purchase feedback and adjust the model as to which recommendations were helpful, which were not, and even which products were harmful to them. This will further promote the highest quality and safest recommendation to users.

The risk of bias in terms of skin tones and skin color also arises in the analysis and modeling phase. Because the skincare industry already highly overrepresented people with lighter skin, the analysis of the model will need to compensate for this to promote fairness and inclusivity of customers with darker skin tones. The model will give additional weight to ratings, comments and feedback from customers with darker skin tones, and will likewise give additional weight to their data regarding how their skin reacted, or didn't react, to their products (e.g. a positive effect, neutral effect, or negative effect). The model will also account for common skincare myths, such as the myth that darker skinned people do not need sunscreen. Dermatologists recognize that all skin tones are susceptible to the damaging effects of prolonged sun exposure, such as skin cancer and early skin aging (Patel et al, 2020). Therefore the model will be sure to promote safe and effective facial sunscreens for all customers regardless of skin tone.

This effort to promote the skin health of people of color will increase confidence in customers of color to know that the recommendations are made with their needs in mind.

Another ethical concern is regarding user transparency and explainability. Users will likely want to understand why certain products are being recommended to them, so that they can have more control over their choices. Customers will want to know what types of ingredients are beneficial to them, enabling them to compare similar products both on the company's website and elsewhere. Lack of explainability and transparency can lead to suspicion and distrust from users if they don't understand the reasoning behind the recommendations, leading to reduced engagement and fewer sales. This is two-fold, as fewer sales will result in fewer insights from customer feedback, which decreases the model's effectiveness. This can be promoted by using interpretability techniques in the analysis phase.

This can also be mitigated by providing clear, understandable notes for each recommendation (e.g., "Product A is recommended because it contains retinol, which is effective for treating wrinkles", or "Product B is recommended because it has bakuchiol, which is effective for treating wrinkles, and is less prone to aggravating rosacea) (Santilli & Jacoby, 2022). As described before, the model should also clearly explain key ingredients and their benefits or potential risks, tailored to the user's specific skin type and concerns.

Use

This model is specifically designed for websites or apps where users purchase skincare products. The model's goal is to enhance customer experience by providing

personalized product recommendations based on skin type, concerns, consumer data patterns, and expert opinions. This model can be implemented by websites that exclusively sell their own brand, as well as those that offer a variety of skincare brands.

The model is not intended to substitute a dermatologist's role in diagnosis and treatment for skin conditions. The model is not intended to prescribe prescription medications such as topical tretinoin cream or oral isotretinoin (Accutane), which are popular skincare medications available only by prescription in the US. The model is also not equipped for situations requiring real-time adjustments, such as severe allergic reactions or other medical reactions. The model is also not designed to be used in contexts where data privacy cannot be assured.

To address the proper interpretation of the outputs of the model in a context-sensitive manner, it will be important for the company to practice transparency regarding the results. Disclosures should be made readily available and readable to the user upon the presentation of the product recommendations, which states: This is only a predictive analysis and any concerns or questions should be directed to a dermatologist prior to usage of the product(s). For each product recommended, another disclaimer should be included as to which ingredients may have harmful effects if mixed with other ingredients, and on which skin tones the ingredients have been shown to have negative/negligent effects. The company can further enhance this by offering access to a staffed dermatologist or qualified esthetician to further answer customers' questions via a web chat service.

Justice

Proper use of this model can result in increased justice due to the enhanced access to skincare expertise. In the US specifically, healthcare related to usually non-urgent matters such as longterm skin health and cosmetic appearance are usually not highly prioritized by the general public due to the expensive nature of medical healthcare. People may feel that they need to devote their time, energy, and money into managing more urgent or critical healthcare concerns. The complex and often trial-by-error nature of skincare means that, traditionally, only people of higher socioeconomic status have the means to become familiar with the endless options of skincare ingredients and product types, or to hire a professional to handle this endeavor for them. Additionally, individuals in underserved or rural areas often have limited dermatological/esthetician services nearby and available. The model will provide expert-validated skincare recommendations to the general public who desire to improve their skin health and appearance, thereby improving their psychological wellbeing. Consumers will also save time and money by avoiding skincare products which are not likely to suit their needs.

Proper use of this model also increases justice by empowering consumers with educational insights. In the transparency process, educating users about suitable skincare products and ingredients for their specific needs and rationales for why specific recommendations were chosen leads to better-informed purchasing decisions and improved skincare outcomes. With increased knowledge about skincare science, consumers will be empowered to make educated choices for themselves in regards to the health of their skin.

Proper use of this model also increases justice by addressing underrepresented demographics in skincare, namely people with darker skin tones. This model highlights and addresses gaps in the current skincare market by compensating for prior bias and myths to produce quality skincare recommendations for all skin tones. In doing so, this also likely promotes diverse skincare brands, including those from minority-owned businesses, promoting economic justice and diversity in the skincare market.

Revision

The model will benefit from regular periodic monitoring for effectiveness and regular audits for bias. It is also important to consider specific potential situations which call for the update or revision of the model for ethical reasons. One such reason is emerging ingredients and further skincare research. For example, new ingredients coming into the skincare market, or additional risks and benefits of current skincare ingredients will come to light with ongoing dermatological research. New skincare trends will also change the popularity of certain ingredients or products over others. Therefore, the expert review guidance portion of the model should be updated and revised to account for the change in information available as well as changing trends. This will be crucial to continue to maintain the trust of users and to avoid harmful effects on users. By addressing these aspects, the machine learning model can be developed and deployed ethically by ensuring it serves its intended purpose effectively and fairly.

Another potential situation is the generalization of the model to the makeup industry. Skincare and makeup share several commonalities in that they are both applied to the skin/face, both have a wide variety of ingredients and product types, and

both would benefit from a model to assist customers in finding the right selection.

However, significant reworkings of the model will need to be accomplished in order to provide quality recommendations for customers. The model would need to collect new data specific to makeup products, including user reviews, expert opinions, and ingredient lists. It would also need to rework the product categories from cleansers, toners, etc to fit an even wider range of makeup categories (e.g., foundation, concealer, bronzer, highlight, eyeshadow, mascara, lipstick, lip gloss, lip stain, etc). The model may also need to develop subcategories to reflect the style of the look (matte, dewey, shimmer, etc). Whereas this current model only accounts for skin tones of three categories (light, medium, dark), a makeup model would need to greatly expand this into many shades of skin, as well as account for warmth/coolness/olive tones, to recommend suitable skin-specific products such as foundations and concealers (Jilek, 2022). By carefully providing these adjustments, the transition from a skincare selection model to a makeup selection model can be made ethically in order to ensure that users receive personalized, trustworthy, and inclusive recommendations.

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